Lecture 3: Practice Problems

Math for Machine Learning

October 26, 2026

1 Multivariable Functions

Problem 1.1 (**Mandatory**)**.** For each function, find the largest open set on which it is defined, describe its level sets, and sketch a contour plot.

(a) $f(x, y) = 2x - y + 1$;

(b) $f(x, y) = 4x^2 + y^2$;

(c) $f(x, y) = y - x^2$;

(d) $f(x, y) = \log(x^2 + y^2)$.

Problem 1.2 ($\star$)**.** Decide whether the limit as $(x, y) \to (0, 0)$ exists, and find it if it does.

(a) $\dfrac{x^2y^2}{x^2 + y^2}$;

(b) $\dfrac{x^2 - y^2}{x^2 + y^2}$;

(c) $\dfrac{x^3 + y^3}{x^2 + y^2}$;

(d) $\dfrac{xy}{\sqrt{x^2 + y^2}}$.

Problem 1.3 ($\star$)**.** Let $f(x, y) = \frac{xy^2}{x^2+y^4}$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$. Show that $f(t\mathbf{v}) \to 0$ as $t \to 0$ for every vector $\mathbf{v} \in \mathbb{R}^2$, but that f is not continuous at the origin.

Problem 1.4. We fit a linear model $\hat{y} = wx + b$ to the three data points $(1, 2)$, $(2, 3)$ and $(3, 5)$, with the mean squared error

$$L(w, b) = \frac{1}{3}\sum_{i=1}^{3}(wx_i + b - y_i)^2.$$

(a) Write L as a polynomial of degree 2 in w and b.

(b) Explain why the level sets $L(w, b) = c$ are ellipses (for c larger than the minimum value). (Hint: the quadratic part is $\mathbf{p}^\top A\mathbf{p}$ with $\mathbf{p} = (w, b)^\top$; use the section on the spectral theorem in Lecture 2.) We will find the center of these ellipses in Problem 9.4.

2 Derivatives of Functions of One Variable

Problem 2.1 (**Mandatory**)**.** Let $\sigma(x) = \frac{1}{1+e^{-x}}$ be the sigmoid function.

(a) Show that the SiLU (or swish) activation function $f(x) = x\sigma(x)$ has the derivative $f'(x) = \sigma(x)(1 + x(1 - \sigma(x)))$.

(b) Show that $\frac{d}{dx} \log \sigma(x) = 1 - \sigma(x) = \sigma(-x)$.

(c) The ELU activation function is $f(x) = x$ for $x \geq 0$ and $f(x) = e^x - 1$ for $x < 0$. Show that f is differentiable at 0, and find f'. Is f' differentiable at 0?

(d) Compute the one-sided difference quotients of the leaky ReLU $f(x) = \max(0.1x, x)$ at 0. Is it differentiable at 0?

Problem 2.2 (**Mandatory**)**.** Let $f(x) = \sqrt{1 + x}$.

(a) Compute the Taylor polynomial T_3 of f at 0.

(b) Use it to approximate $\sqrt{1.1}$.

(c) Compute $f^{(4)}$, and use Taylor's theorem to show that the error of the approximation in (b) is less than $4 \cdot 10^{-6}$.

Problem 2.3 ($\star$)**.** Use Taylor polynomials at 0 to compute the limits as $x \to 0$:

(a) $\dfrac{\sin x - x}{x^3}$;

(b) $\dfrac{e^x - 1 - x}{x^2}$;

(c) $\dfrac{1 - \cos x}{x^2}$;

(d) $\dfrac{\log(1 + x) - x}{x^2}$.

Problem 2.4 ($\star$)**.** Let f be differentiable on the interval $(-1, 4)$, with $f(3) = 5$ and $f'(x) \geq -1$ for all x. What is the largest possible value of $f(0)$? Find a function f for which it is attained. (Hint: the mean value theorem, i.e., Taylor's theorem with $k = 0$.)

Problem 2.5. Let $f(x) = e^x$ and $x = 0$, so that $f'(0) = 1$.

(a) For $h = 0.1$, compute the forward difference $\frac{f(h)-f(0)}{h}$ and the central difference $\frac{f(h)-f(-h)}{2h}$, and their errors. Compare the errors with the bounds of the proposition on finite differences in the lecture.

(b) Repeat for $h = 0.05$. By which factors do the two errors shrink, and why?

3 Partial Derivatives and Gradients

Problem 3.1 (Mandatory)**.** Compute the gradients of the following functions.

(a) $f(x, y) = x^2y^3 - 4xy$, at the point $(1, -1)$;

(b) $f(\mathbf{x}) = \log(x_1^2 + x_2^2 + x_3^2)$, at the point $(1, 2, 2)$;

(c) $f(\mathbf{x}) = (\mathbf{a}^\top \mathbf{x})^2$, for a fixed vector $\mathbf{a} \in \mathbb{R}^n$;

(d) $f(\mathbf{x}) = e^{-\|\mathbf{x}\|^2/2}$;

(e) $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top \mathbf{x}$, for a symmetric matrix A.

Problem 3.2 (Mandatory)**.** Let $f(x, y) = 3x^2y - y^3$ and $P = (1, 2)$.

(a) Compute $\nabla f(P)$.

(b) Compute the directional derivative of f at P in the direction towards the point $Q = (4, 6)$.

(c) In which directions does f increase and decrease fastest at P, and at which rates?

(d) Find all unit vectors $\mathbf{v}$ with $D_{\mathbf{v}} f(P) = 0$.

Problem 3.3 ($\star$)**.** The temperature on a metal plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at the point $(2, 1)$ wants to get warm as quickly as possible.

(a) In which direction should it start to move, and at which rate does the temperature increase initially?

(b) Suppose the bug always moves in the direction of ∇T. Show that it moves along the parabola $y = \frac{x^2}{4}$, by checking that the tangent vectors of this parabola are parallel to ∇T. Where does the bug end up?

Problem 3.4. Let $f(x, y) = \frac{x^2 y}{x^2+y^2}$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$.

(a) Show that f is continuous at the origin. (Hint: $|f(x, y)| \leq |y|$.)

(b) Compute the directional derivative $D_{\mathbf{v}} f(0, 0)$ for every unit vector $\mathbf{v} = (a, b)^\top$.

(c) Conclude that f is not differentiable at the origin, although it is continuous and all its directional derivatives exist there.

Problem 3.5. Let $A = \mathrm{diag}(1, 4)$, with the inner product $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A\mathbf{y}$ and the norm $\|\mathbf{x}\|_A = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_A}$, and let $f(\mathbf{x}) = x_1^2 + x_2^2$.

(a) Find the vector $\mathbf{g}$ with $Df(1, 1)\mathbf{h} = \langle \mathbf{g}, \mathbf{h} \rangle_A$ for all $\mathbf{h} \in \mathbb{R}^2$, i.e., the gradient of f at $(1, 1)$ with respect to this inner product.

(b) Show that among all $\mathbf{h}$ with $\|\mathbf{h}\|_A = 1$, the rate of change $Df(1, 1)\mathbf{h}$ is largest for $\mathbf{h} = \mathbf{g}/\|\mathbf{g}\|_A$.

(c) Compare $\mathbf{g}$ with the usual gradient $\nabla f(1, 1)$. Why are they different?

4 Jacobians: Derivatives of Vector-Valued Functions

Problem 4.1 (**Mandatory**, $\star$)**.** Let $F(x, y) = (x^2 - y^2,\, 2xy)^\top$. (If we identify (x, y) with the complex number $z = x + iy$, then F is the map $z \mapsto z^2$.)

(a) Compute the Jacobian matrix and the Jacobian determinant of F.

(b) Use the linear approximation at $(1, 2)$ to approximate $F(1.1, 1.9)$, and compare with the exact value.

(c) By which factor does F scale small areas near (x, y)? Where does F preserve areas locally?

Problem 4.2. Compute the Jacobians of the following functions.

(a) ReLU applied entrywise to $\mathbf{x} \in \mathbb{R}^n$, at a point $\mathbf{x}$ with no zero entries.

(b) The normalization $f(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$ for $\mathbf{x} \neq \mathbf{0}$. Show that $Df(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|}(I - \mathbf{u}\mathbf{u}^\top)$ with $\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}$, and that $Df(\mathbf{x})\mathbf{x} = \mathbf{0}$. Why should we expect the last equation?

(c) The softmax function with temperature $T > 0$, $f(\mathbf{z}) = \mathrm{softmax}(\mathbf{z}/T)$.

Problem 4.3 ($\star$)**.** Use polar coordinates to compute:

(a) $\iint_{x^2+y^2 \le 1} (x^2 + y^2)\, dx\, dy$;

(b) $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\, dx\, dy$;

(c) the probability that a standard normal random vector in $\mathbb{R}^2$, with the density $\frac{1}{2\pi} e^{-(x^2+y^2)/2}$, lies in the disk of radius r around the origin. For which r is this probability 95%? (Compare with the number 1.96 for one standard normal variable.)

5 The Chain Rule

Problem 5.1 (**Mandatory**)**.** Let $g(x, y) = (x + y,\, xy)^\top$ and $f(u, v) = u^2 - 2v$. Compute $D(f \circ g)(x, y)$ in two ways: as the product $Df(g(x, y))\, Dg(x, y)$ of the Jacobians, and by simplifying $f(g(x, y))$ first.

Problem 5.2 (**Mandatory**)**.** Use the chain rule to compute the following gradients, and write down the sizes of all matrices in your computation.

(a) The logistic loss with labels $y \in \{+1, -1\}$: $\ell(\mathbf{w}) = \log\left(1 + e^{-y\,\mathbf{w}^\top \mathbf{x}}\right)$.

(b) Ridge regression: $f(\mathbf{w}) = \frac{1}{2}\|X\mathbf{w} - \mathbf{y}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, with $X \in \mathbb{R}^{N \times d}$.

(c) The (unsquared) residual norm $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|$, at points with $A\mathbf{x} \neq \mathbf{b}$.

Problem 5.3. Let $T(x, y) = x^2 y$, and let $\mathbf{x}(t) = (\cos t, \sin t)^\top$ move on the unit circle.

(a) Compute $\frac{d}{dt} T(\mathbf{x}(t))$ with the chain rule, and check your answer by differentiating $T(\mathbf{x}(t))$ directly. Evaluate it at $t = \frac{\pi}{4}$.

(b) Find all t with $\frac{d}{dt} T(\mathbf{x}(t)) = 0$, and the maximum of T on the unit circle.

Problem 5.4 ($\star$)**.** Use the fact that gradients are orthogonal to level sets to find:

(a) the tangent line to the ellipse $x^2 + 4y^2 = 8$ at the point $(2, 1)$;

(b) the tangent plane to the sphere $x^2 + y^2 + z^2 = 9$ at the point $(1, 2, 2)$;

(c) the tangent plane to the graph of $z = x^2 + y^2$ at the point $(1, 2, 5)$. (Hint: the graph is a level set of $F(x, y, z) = x^2 + y^2 - z$.)

Problem 5.5 ($\star$)**.** The equation $x^2 + xy + y^3 = 3$ defines y as a differentiable function $y(x)$ of x near the point $(1, 1)$. Differentiate both sides of $x^2 + x\,y(x) + y(x)^3 = 3$ with the chain rule to find $y'(1)$. Show more generally that if $F(x, y(x)) = c$, then $y'(x) = -\frac{\partial F/\partial x}{\partial F/\partial y}$, wherever $\frac{\partial F}{\partial y} \neq 0$.

6 Gradients with Respect to Matrices

Problem 6.1 (**Mandatory**)**.** Use first-order expansions $f(X + H) = f(X) + \langle G, H\rangle + o(\|H\|_F)$ to show:

(a) $\nabla_X \operatorname{tr}(AX^\top) = A$ for $A, X \in \mathbb{R}^{m\times n}$;

(b) $\nabla_X \|X\mathbf{a}\|^2 = 2X\mathbf{a}\mathbf{a}^\top$ for $\mathbf{a} \in \mathbb{R}^n$;

(c) $\nabla_X \operatorname{tr}(X^2) = 2X^\top$ for $X \in \mathbb{R}^{n\times n}$;

(d) $\nabla_X \|X\|_F = \frac{X}{\|X\|_F}$ for $X \neq 0$.

Problem 6.2 (**Mandatory**)**.** Let $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^m$ and $\lambda > 0$, and let $L(W) = \frac{1}{2}\|W\mathbf{x} - \mathbf{y}\|^2 + \frac{\lambda}{2}\|W\|_F^2$ for $W \in \mathbb{R}^{m\times n}$.

(a) Show that $\nabla_W L = (W\mathbf{x} - \mathbf{y})\mathbf{x}^\top + \lambda W$.

(b) Show that $W^* = \frac{\mathbf{y}\mathbf{x}^\top}{\|\mathbf{x}\|^2+\lambda}$ is a critical point. (It is the unique minimum, since L is strictly convex; see the section on convexity in the lecture.)

(c) Compute $W^*\mathbf{x}$. What happens as $\lambda \to 0$, and as $\lambda \to \infty$?

Problem 6.3. (a) For $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, compute the four partial derivatives of $\det(X) = ad - bc$, and check that $\nabla \det(X) = \det(X)X^{-\top}$ if X is invertible.

(b) Let X be symmetric positive definite. Show that $\nabla_X \log\det(X) = X^{-1}$.

Problem 6.4. A classifier with three classes computes the scores $\mathbf{z} = (1, 2, 3)^\top$, and the correct class is $c = 3$.

(a) Compute the predicted probabilities $\mathbf{p} = \operatorname{softmax}(\mathbf{z})$, the cross-entropy loss $\ell = -\log p_3$, and $\nabla_{\mathbf{z}}\ell$.

(b) With a temperature $T > 0$, the probabilities are $\mathbf{p} = \operatorname{softmax}(\mathbf{z}/T)$. Show that then $\nabla_{\mathbf{z}}\ell = \frac{1}{T}(\mathbf{p} - \mathbf{e}_3)$, and evaluate the loss and the gradient for $T = \frac{1}{2}$ and $T = 2$.

7 Backpropagation and Automatic Differentiation

Problem 7.1. Let $f(x, y) = (x + y)\, e^{xy}$. Draw the computation graph of f with the intermediate variables $v_1 = x + y$, $v_2 = xy$, $v_3 = e^{v_2}$ and $v_4 = v_1 v_3$. Compute $\nabla f(2, 0)$ in forward mode (two passes) and in reverse mode (one pass), and check the result by differentiating f directly.

Problem 7.2 (Mandatory)**.** Consider the network $\mathbf{z} = W_1\mathbf{x} + \mathbf{b}_1$, $\mathbf{a} = \mathrm{ReLU}(\mathbf{z})$, $\hat{y} = \mathbf{w}_2^\top \mathbf{a} + b_2$ with the loss $L = \frac{1}{2}(\hat{y} - y)^2$, and the values

$$W_1 = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{b}_1 = \mathbf{0}, \quad \mathbf{w}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad b_2 = 0.5, \quad \mathbf{x} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}, \quad y = 1.5.$$

(a) Carry out the forward pass, and compute L.

(b) Carry out the backward pass: compute $\nabla_{\mathbf{w}_2} L$, $\frac{\partial L}{\partial b_2}$, $\nabla_{\mathbf{a}} L$, $\nabla_{\mathbf{z}} L$, $\nabla_{W_1} L$ and $\nabla_{\mathbf{b}_1} L$.

(c) Which weights receive a zero gradient, and why? What would happen to them in gradient descent if this were the only training example?

Problem 7.3. (a) How many passes of forward mode and how many passes of reverse mode are needed to compute the full Jacobian of a function $f : \mathbb{R}^{1000} \to \mathbb{R}$? Of a function $f : \mathbb{R} \to \mathbb{R}^{1000}$?

(b) Let $f = f_3 \circ f_2 \circ f_1$ with Jacobians $J_1 \in \mathbb{R}^{100 \times 10^4}$, $J_2 \in \mathbb{R}^{100 \times 100}$ and $J_3 \in \mathbb{R}^{1 \times 100}$. Count the multiplications needed to compute $J_3 J_2 J_1$ as $J_3(J_2 J_1)$ and as $(J_3 J_2) J_1$. Which order corresponds to reverse mode?

Problem 7.4 ($\star$)**.** Let $f_L(x) = \sigma(\sigma(\cdots \sigma(x) \cdots))$ be the L-fold composition of the sigmoid function.

(a) Show that $|f_L'(x)| \le 4^{-L}$ for all x. How small is this bound for $L = 10$?

(b) Now let every layer have a weight w, i.e., $g_1(x) = \sigma(wx)$ and $g_{l+1}(x) = \sigma(w\, g_l(x))$. Show that $|g_L'(x)| \le (\frac{|w|}{4})^L$. What does this suggest about the size of the weights in deep sigmoid networks?

8 Higher Derivatives and Taylor's Theorem

Problem 8.1 (Mandatory)**.** Let $f(x, y) = x^3 - 2xy^2 + y^4$.

(a) Compute ∇f and $\nabla^2 f$, and check that the mixed partial derivatives are equal.

(b) Compute the second-order Taylor approximation of f at the point $(1, 1)$. Use it to approximate $f(1.1, 0.9)$, and compare with the exact value.

(c) Compute the curvature $\mathbf{v}^\top \nabla^2 f(1, 1)\mathbf{v}$ of f at $(1, 1)$ in the direction $\mathbf{v} = \frac{1}{\sqrt{2}}(1, 1)^\top$. What are the largest and the smallest curvature at $(1, 1)$ over all unit directions?

Problem 8.2. Compute the Hessians of the following functions.

(a) Ridge regression: $f(\mathbf{w}) = \frac{1}{2}\|X\mathbf{w} - \mathbf{y}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$.

(b) $f(\mathbf{x}) = \|\mathbf{x}\|$ for $\mathbf{x} \neq \mathbf{0}$. Find the eigenvalues and eigenvectors of the Hessian. Why is one eigenvalue 0?

(c) $f(\mathbf{w}) = \log(1 + e^{\mathbf{w}^\top \mathbf{x}})$. What is the rank of the Hessian?

Problem 8.3. Let $f(x, y) = \log(e^x + e^y)$, the log-sum-exp function in two variables.

(a) Compute $\nabla f(0, 0)$ and $\nabla^2 f(0, 0)$.

(b) Use the second-order Taylor approximation at the origin to approximate $f(0.2, -0.1)$, and compare with the exact value.

(c) Find the eigenvalues and eigenvectors of $\nabla^2 f(0, 0)$. Explain the eigenvalue 0 using the identity $f(x + c, y + c) = f(x, y) + c$.

Problem 8.4 ($\star$)**.** Let $f(x, y) = \frac{xy(x^2-y^2)}{x^2+y^2}$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$.

(a) Show that $\frac{\partial f}{\partial x}(0, y) = -y$ and $\frac{\partial f}{\partial y}(x, 0) = x$ for all x and y.

(b) Conclude that $\frac{\partial^2 f}{\partial y \partial x}(0, 0) = -1$, but $\frac{\partial^2 f}{\partial x \partial y}(0, 0) = 1$. Why does this not contradict Schwarz's theorem?

9 Unconstrained Optimization

Problem 9.1 (**Mandatory**)**.** Find all critical points of the following functions, and classify them with the eigenvalues of the Hessian. Which local minima are global minima?

(a) $f(x, y) = x^2 + xy + y^2 - 3x$;

(b) $f(x, y) = x^3 - 3x + y^2 - 2y$.

Problem 9.2 ($\star$)**.** Let $f(x, y) = xy(x + y - 3)$.

(a) Find all four critical points of f, and classify them.

(b) The level set $f = 0$ consists of three lines. Sketch them, and mark the regions where $f > 0$ and $f < 0$. Explain the types of the critical points from this picture.

Problem 9.3 (**Mandatory**)**.** (a) Let $f(x, y) = 3x^2 + 2xy + 3y^2$. Find the eigenvalues and eigenvectors of the Hessian at the critical point $(0, 0)$. Describe and sketch the level curves of f: in which direction are they longest, and what is the ratio of their axes?

(b) Let $g(x, y) = x^2 + 6xy + y^2$. Find the eigenvalues and eigenvectors of the Hessian at the origin, and classify the critical point. Along which lines through the origin does g increase, and along which does it decrease? Find the level set $g = 0$, and sketch a contour plot of g.

Problem 9.4 (**Mandatory**)**.** Let $L(w, b)$ be the mean squared error of Problem 1.4.

(a) Compute ∇L, and solve $\nabla L = \mathbf{0}$.

(b) Compute the Hessian of L and its eigenvalues, and explain why the critical point is the global minimum.

(c) Compute the minimum value of L, and check that the solution satisfies the normal equations of Lecture 2.

Problem 9.5 ($\star$)**.** Let $f(x, y) = (y - x^2)(y - 2x^2)$.

(a) Show that the origin is a critical point, and that the second-order test is inconclusive there.

(b) Show that on every line through the origin, the restriction of f has a strict local minimum at the origin.

(c) Show that the origin is nevertheless not a local minimum of f. (Hint: look at the parabola $y = \frac{3}{2}x^2$.)

10 Gradient Descent and Newton's Method

Problem 10.1 (**Mandatory**)**.** We fit the model $\hat{y} = wx$ to the data $(1, 1)$, $(2, 2)$, $(3, 2)$ by minimizing the mean squared error $L(w) = \frac{1}{3}\sum_{i=1}^{3}(wx_i - y_i)^2$.

(a) Derive the gradient descent update $w_{k+1} = w_k - \eta L'(w_k)$ explicitly, and find the minimum w^*.

(b) Carry out three steps with $\eta = 0.05$, starting at $w_0 = 0$. By which factor does the error $w_k - w^*$ shrink in every step?

(c) For which η does gradient descent converge? Which η reaches w^* in a single step?

(d) Write down the update for general data $\mathbf{x}_i \in \mathbb{R}^d$ in vector form, and explain how each data point influences the step.

Problem 10.2 (**Mandatory**)**.** Let $f(x, y) = 2x^2 + 8y^2$.

(a) Carry out two steps of gradient descent with step size $\eta = 0.1$, starting at $(1, 1)$.

(b) For which step sizes does gradient descent converge for every starting point? Find the best step size and the corresponding rate ρ.

(c) How many steps with the best step size are needed to reduce the error by a factor 10^{-6}?

(d) What happens for $\eta = 0.125$?

Problem 10.3. Consider again the mean squared error $L(w, b)$ of Problems 1.4 and 9.4.

(a) Which step sizes make gradient descent converge? Find the best step size, the rate ρ, and the number of steps needed to reduce the error by a factor 10^{-6}.

(b) Now center the feature: use the model $\hat{y} = w(x - 2) + c$, where 2 is the mean of the x_i. Compute the new Hessian, its condition number, the best rate, and the number of steps. How do you get b back from c?

(c) Explain the improvement in terms of the columns of the design matrix.

Problem 10.4 (**Mandatory**, $\star$)**.** Let $f(x) = x - \log x$ for $x > 0$.

(a) Show that f has its global minimum at $x^* = 1$.

(b) Show that Newton's method is $x_{k+1} = 2x_k - x_k^2$, and that $1 - x_{k+1} = (1 - x_k)^2$. Compute the iterates for $x_0 = 0.5$.

(c) For which starting points $x_0 > 0$ does Newton's method converge? What happens for $x_0 = 3$?

Problem 10.5 ($\star$)**.** Show that Newton's method for minimizing $f(x) = \frac{x^3}{3} - 2x$ on $(0, \infty)$ is $x_{k+1} = \frac{1}{2}(x_k + \frac{2}{x_k})$, and compute three iterates starting at $x_0 = 1$. Which number does it compute? How would you compute $\sqrt{a}$ for $a > 0$ in the same way? (This method was already known in ancient Babylon.)

Problem 10.6. Let $s(x) = \log(1 + e^x)$ be the softplus function.

(a) Show that $0 < s''(x) \leq \frac{1}{4}$, so that s is $\frac{1}{4}$-smooth.

(b) What does the theorem on gradient descent for smooth functions guarantee for gradient descent on s with $\eta = 4$? The function s has no minimum. What do the iterates do, and why is this consistent with the theorem?

11 Convexity

Problem 11.1 (**Mandatory**)**.** Which of the following functions are convex? Justify your answers.

(a) $x^2 + xy + y^2$ on $\mathbb{R}^2$;

(b) $x^2 + 4xy + y^2$ on $\mathbb{R}^2$;

(c) $e^{\|\mathbf{x}\|^2}$ on $\mathbb{R}^n$;

(d) $\|\mathbf{x}\|_\infty$ on $\mathbb{R}^n$;

(e) $-\sqrt{x}$ on $(0, \infty)$;

(f) $\frac{x^2}{y}$ on $\{(x, y) : y > 0\}$;

(g) $\frac{x}{y}$ on $\{(x, y) : y > 0\}$.

Problem 11.2 ($\star$)**.** (a) Show that log is concave on $(0, \infty)$.

(b) Deduce that $a^\theta b^{1-\theta} \leq \theta a + (1 - \theta)b$ for all $a, b > 0$ and $\theta \in [0, 1]$. In particular, $\sqrt{ab} \leq \frac{a+b}{2}$.

(c) Find the minimum of $x + \frac{4}{x}$ on $(0, \infty)$ in two ways: with (b), and with derivatives and convexity.

Problem 11.3 (**Mandatory**)**.** (a) Show that the logistic loss $\ell(\mathbf{w}) = \log\left(1 + e^{-y\,\mathbf{w}^\top \mathbf{x}}\right)$ with $y \in \{\pm 1\}$ is convex, by computing its Hessian (use Problem 5.2).

(b) Show that the hinge loss $\max(0, 1 - y\,\mathbf{w}^\top \mathbf{x})$ is convex.

(c) Show that the lasso objective $\|X\mathbf{w} - \mathbf{y}\|^2 + \lambda\|\mathbf{w}\|_1$ with $\lambda \geq 0$ is convex.

(d) The "ℓ_0 norm" $\|\mathbf{w}\|_0$ is the number of nonzero entries of $\mathbf{w}$. Show that it is not convex, already for $n = 1$.

Problem 11.4. Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable, and let $C \subseteq \mathbb{R}^n$ be convex. Show that if $\mathbf{x}^* \in C$ satisfies $\nabla f(\mathbf{x}^*)^\top(\mathbf{y} - \mathbf{x}^*) \geq 0$ for all $\mathbf{y} \in C$, then $\mathbf{x}^*$ is a minimum of f on C. Draw a picture for a half-plane C and a point $\mathbf{x}^*$ on its boundary.

Problem 11.5 ($\star$)**.** The simplest "deep" network has two scalar weights: $\hat{y} = w_2 w_1 x$. For the single training example $x = y = 1$, the squared loss is $f(w_1, w_2) = (w_1 w_2 - 1)^2$.

(a) Find all global minima of f.

(b) Show that f is not convex.

(c) Find all critical points, and classify the ones that are not global minima. What are the eigenvalues of the Hessian at the minimum $(1, 1)$, and why is one of them 0?

12 Constrained Optimization

Problem 12.1 (Mandatory)**.** Find the maximum and the minimum of $f(x, y) = xy$ on the ellipse $x^2 + 4y^2 = 8$, using a Lagrange multiplier. Sketch the ellipse and some level curves of f. Check that at the maximum $(2, 1)$, the level curve $xy = 2$ and the ellipse have the same tangent line (compare with Problem 5.4).

Problem 12.2 ($\star$)**.** (a) Find the maximum of xyz subject to $x + y + z = 12$ and $x, y, z > 0$.

(b) An open box (without a lid) with sides $x, y, z > 0$ (where z is the height) has volume 32. Find the dimensions that minimize its surface area $xy + 2xz + 2yz$.

Problem 12.3. Find the points of the ellipse $x^2 + xy + y^2 = 3$ that are closest to and farthest from the origin. (Hint: the Lagrange condition says that $\mathbf{x}$ is an eigenvector of a symmetric matrix.)

Problem 12.4. Find the solution of the system $x_1 + x_2 + x_3 = 3$, $x_1 - x_3 = 1$ with the smallest norm, using the formula $\mathbf{x}^* = A^\top (AA^\top)^{-1}\mathbf{b}$ from the lecture. Find a vector $\mathbf{v}$ that spans the null space of the coefficient matrix, check that $\mathbf{x}^* \perp \mathbf{v}$, and compute $\|\mathbf{x}^* + t\mathbf{v}\|^2$ for all $t \in \mathbb{R}$.

Problem 12.5 (Mandatory)**.** (a) Minimize $(x - 3)^2 + (y - 2)^2$ subject to $x + y \leq 2$, using the KKT conditions. Interpret the result as a projection, and compare the minimal distance with the formula for the distance to a hyperplane in Lecture 2.

(b) Minimize $x^2 + (y - 5)^2$ subject to $x + y \leq 2$, $x \geq 0$ and $y \geq 0$. Which constraints are active, and what are the multipliers?

Problem 12.6 ($\star$)**.** Find the rectangular box with sides parallel to the coordinate axes and the largest volume that fits into the unit ball $x^2 + y^2 + z^2 \leq 1$. Which fraction of the volume of the ball does it fill?

13 Vector Fields: Divergence and Curl

Problem 13.1 (Mandatory)**.** Compute the divergence and the curl of the following vector fields. Which of them are gradient fields? For those, find a function f with $F = \nabla f$.

(a) $F(x, y) = (x^2 - y^2,\, 2xy)^\top$;

(b) $F(x, y) = (2x + y,\, x - 3y)^\top$;

(c) $F(x, y) = (y\cos(xy),\, x\cos(xy))^\top$;

(d) $F(x, y, z) = (y^2,\, 2xy + z,\, y)^\top$.

Problem 13.2 (Mandatory)**.** (a) Compute the Laplacians of $x^2 + y^2 - 2z^2$, of $e^x \sin y$, and of $\|\mathbf{x}\|^2$ on $\mathbb{R}^n$.

(b) Show that at a critical point of a harmonic function on $\mathbb{R}^2$, the Hessian is never positive definite. (So a harmonic function has no strict local minimum of this kind.)

(c) Let p be the density of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, i.e., $\log p(\mathbf{x}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) + c$. Compute the score $\mathbf{s} = \nabla \log p$ and its divergence $\Delta \log p$. Evaluate the divergence for $\Sigma = \operatorname{diag}(1, 4)$.

Problem 13.3. Split $A = \begin{pmatrix} 1 & 3 \\ -1 & 1 \end{pmatrix}$ into its symmetric part S and its skew-symmetric part W. Describe how the vector field $F(\mathbf{x}) = A\mathbf{x}$ acts on a small neighborhood of the origin: along which directions does it stretch, in which direction does it rotate, and how fast? Compute $\operatorname{div} F$ and $\operatorname{curl} F$.

Problem 13.4 ($\star$)**.** (Gradient descent–ascent.) Consider the game $\min_x \max_y xy$.

(a) Show that simultaneous gradient descent in x and ascent in y moves in the direction of the vector field $F(x, y) = (-y, x)^\top$. Compute its divergence and curl, and conclude that F is not of the form $-\nabla g$ for any function g: there is no loss that gradient descent–ascent minimizes.

(b) Show that the simultaneous update $(x, y) \leftarrow (x - \eta y,\ y + \eta x)$ multiplies $x^2 + y^2$ by $1 + \eta^2$.

(c) For the alternating update $x \leftarrow x - \eta y$, then $y \leftarrow y + \eta x$ (with the new x), show that $Q(x, y) = x^2 - \eta xy + y^2$ does not change. Conclude that the iterates stay bounded for $0 < \eta < 2$.

14 Multiple Integrals and the Divergence Theorem

Problem 14.1 (**Mandatory**)**.** (a) Compute $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$. (Hint: e^{x^2} has no elementary antiderivative. Sketch the region and change the order of integration.)

(b) Compute $\iint_D xy\, dx\, dy$ for the quarter disk $D = \{(x, y) : x, y \geq 0,\ x^2 + y^2 \leq 1\}$ in polar coordinates.

(c) Compute $\iint_{[0,1]^2} (x + y)^2\, dx\, dy$.

Problem 14.2 (**Mandatory**)**.** Let X and Y be independent and uniform on $[0, 1]$, so that (X, Y) has the density 1 on the unit square. Compute as double integrals:

(a) $P(X + Y \leq \frac{1}{2})$;

(b) $P(XY \leq \frac{1}{2})$;

(c) the expected value $\iint \max(x, y)\, dx\, dy$ of $\max(X, Y)$.

Problem 14.3 (**Mandatory**)**.** Let A be symmetric positive definite.

(a) Write $A = B^\top B$ with an invertible B, and substitute $\mathbf{u} = B\mathbf{x}$ to show that the ellipse $\{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}^\top A\mathbf{x} \leq 1\}$ has the area $\frac{\pi}{\sqrt{\det A}}$, and that $\int_{\mathbb{R}^2} e^{-\frac{1}{2}\mathbf{x}^\top A\mathbf{x}}\, d\mathbf{x} = \frac{2\pi}{\sqrt{\det A}}$.

(b) Evaluate both for $\mathbf{x}^\top A\mathbf{x} = 2x^2 + 2xy + y^2$.

Problem 14.4 (**Mandatory**)**.** Let $F(x, y) = (-y, x)^\top$ and $G(x, y) = (y, x)^\top$, and let $\gamma_1(t) = (t, t^2)$ and $\gamma_2(t) = (t, t)$ for $0 \leq t \leq 1$, two curves from $(0, 0)$ to $(1, 1)$.

(a) Compute the line integrals of F and of G along both curves. Explain the results with the curl.

(b) Find f with $G = \nabla f$, and check the fundamental theorem for line integrals.

(c) Going along γ_1 and back along γ_2 is a closed curve. Check Green's theorem for F on the region that it encloses.

Problem 14.5. (Score matching in one dimension.) Data $x_1, \dots, x_N$ come from an unknown density p on $\mathbb{R}$. We model the score by $s_\theta(x) = -\theta x$ with $\theta > 0$, which is the score of $\mathcal{N}(0, \frac{1}{\theta})$.

(a) Show that the score matching loss of the lecture is $J(\theta) = \int p(x)\big(\frac{1}{2}\theta^2 x^2 - \theta\big)\,dx + C$.

(b) Replace the integral by the average over the data, and minimize over θ. Compare the result with the maximum likelihood estimate of the precision $\frac{1}{\sigma^2}$ of $\mathcal{N}(0, \sigma^2)$, which is $\big(\frac{1}{N}\sum_i x_i^2\big)^{-1}$.

(c) Evaluate the estimate for the data $1, -2, 3, 0, -2$.

Problem 14.6. Let X have the density p_θ of $\mathcal{N}(\theta, 1)$, so that $\frac{\partial}{\partial\theta}\log p_\theta(x) = x - \theta$, and let $g(\theta) = \int x^2 p_\theta(x)\,dx = \theta^2 + 1$.

(a) Show that the score function formula of the lecture gives $g'(\theta) = \int x^2(x-\theta)\,p_\theta(x)\,dx$, and check that this equals 2θ. (Substitute $x = \theta + z$.)

(b) The reparameterization $X = \theta + Z$ with $Z \sim \mathcal{N}(0,1)$ gives $g(\theta) = \int(\theta+z)^2\varphi(z)\,dz$, with the standard normal density φ. Differentiate under the integral sign to get a second formula for $g'(\theta)$.

(c) For $\theta = 0$, the two formulas average the single-sample estimates X^3 and $2X$. Compare their mean squares, using that $\int z^6\varphi(z)\,dz = 15$. Which estimator would you use?

Problem 14.7 ($\star$)**.** Let $F(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|^3}$ on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$ (the electric field of a point charge).

(a) Show that $\operatorname{div} F = 0$.

(b) Show that the flux of F out of every sphere $\|\mathbf{x}\| = R$ is 4π. (The sphere has the area $4\pi R^2$.)

(c) Why does this not contradict the divergence theorem? What is the flux out of the region between two spheres around the origin?

15 Ordinary Differential Equations

Problem 15.1 (**Mandatory**)**.** (a) Solve $x' = -2x$ with $x(0) = 3$. Compute $x(1)$ and the time after which x has halved.

(b) Show that $x(t) = \frac{1}{1+4e^{-t}}$ solves the logistic equation $x' = x(1-x)$ with $x(0) = 0.2$, and write it with the sigmoid function. Compute $x(2)$ and $\lim_{t\to\infty} x(t)$.

Problem 15.2. (a) Show that $x(t) = \frac{1}{1-t}$ solves $x' = x^2$ with $x(0) = 1$. What happens as $t \to 1$? Why does this not contradict the existence and uniqueness theorem?

(b) Show that $x(t) = 0$ and $x(t) = t^2$ both solve $x' = 2\sqrt{|x|}$ for $t \geq 0$, with $x(0) = 0$. Which assumption of the theorem fails?

Problem 15.3 (Mandatory). Let $A = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$.

(a) Find the eigenvalues and eigenvectors of A, and solve $\mathbf{x}' = A\mathbf{x}$ with $\mathbf{x}(0) = (1, 0)^\top$. Compute $\mathbf{x}(1)$.

(b) For which initial values does the solution tend to $\mathbf{0}$? Sketch the solutions. (The origin is a saddle point of this ODE.)

(c) Compute e^{At}.

Problem 15.4 ($\star$). Let $A = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$ with $\omega > 0$.

(a) Show that the solutions of $\mathbf{x}' = A\mathbf{x}$ move on circles with angular speed ω. What is the period?

(b) Show that Euler's method satisfies $\|\mathbf{x}_{k+1}\|^2 = (1 + h^2\omega^2)\|\mathbf{x}_k\|^2$, so the numerical solution spirals outwards. For $\omega = 1$ and $h = 0.1$, by which factor does $\|\mathbf{x}_k\|$ grow during one period?

Problem 15.5 (Mandatory). Let $f(x_1, x_2) = x_1^2 + 4x_2^2$.

(a) Solve the gradient flow $\mathbf{x}' = -\nabla f(\mathbf{x})$ with $\mathbf{x}(0) = (2, 1)^\top$, and check that $f(\mathbf{x}(t))$ decreases.

(b) Write down Euler's method for this flow with step h, and compare it with gradient descent. For which h do the iterates converge to $\mathbf{0}$? What happens for $h = 0.2$ and $h = 0.3$?

Problem 15.6. Apply Euler's method to $x' = -x$, $x(0) = 1$ on the interval $[0, 1]$, with $h = 0.25$ and $h = 0.125$. Compare with $x(1) = e^{-1}$. By which factor does the error shrink when h is halved?

16 Flows, Volumes and Densities

Problem 16.1 (Mandatory). (a) For the ODE $\mathbf{x}' = A\mathbf{x}$ with $A = \begin{pmatrix} 1 & 2 \\ 0 & -3 \end{pmatrix}$, compute the divergence of the vector field and the factor by which the flow changes areas in the time t. Check the result with the eigenvalues of A.

(b) Let $F(x, y) = (x - x^3, -y)^\top$. Compute $\operatorname{div} F$. Find the points where $F = \mathbf{0}$, and the divergence there. Where does the flow preserve areas to first order, and where does it compress them?

Problem 16.2 ($\star$). Let $x' = -x$, and let $x(0)$ be random with the standard normal density φ.

(a) Show that $x(t) = e^{-t}x(0)$ has the density $p_t(x) = e^t\varphi(e^t x)$.

(b) Verify the continuity equation $\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x}(p_t F)$ for $F(x) = -x$.

(c) Show that $\log p_t(x(t))$ increases at the rate $1 = -\operatorname{div} F$ along each solution, and explain this with the volume interpretation of the divergence.